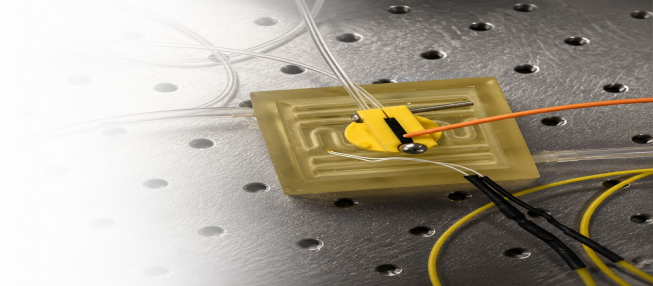


MRI-Compatible Incubation Device for Brain Organoid Imaging

[Project Source](#)

Designed and validated a temperature-controlled incubation system that preserves organoid viability during extended MRI imaging experiments



THE CHALLENGE

High-field MRI provides powerful insights into brain organoid function but current systems cannot maintain organoid viability during long imaging sessions due to lack of temperature control and nutrient delivery

OUR SOLUTION



TEMPERATURE CONTROL

Maintains ~ 37°C with water bath system



MEDIA PERFUSION

Continuous flow for nutrient delivery and waste removal



OPTICAL FIBER PORT

Integrated fiber clamp for optogenetic stimulation

KEY RESULTS



37.3 ± 0.5°C

Stable temperature maintained inside the MR tube



Continuous Perfusion

~400 µL volume maintained with peristaltic pump flow



MRI-Compatible Design

Constructed with non-magnetic materials and optimized for MRI environment



Sustained Metabolic Activity

Metabolic assay shows organoids remain viable after 5 hours of incubation



Preserved Neuronal Activity

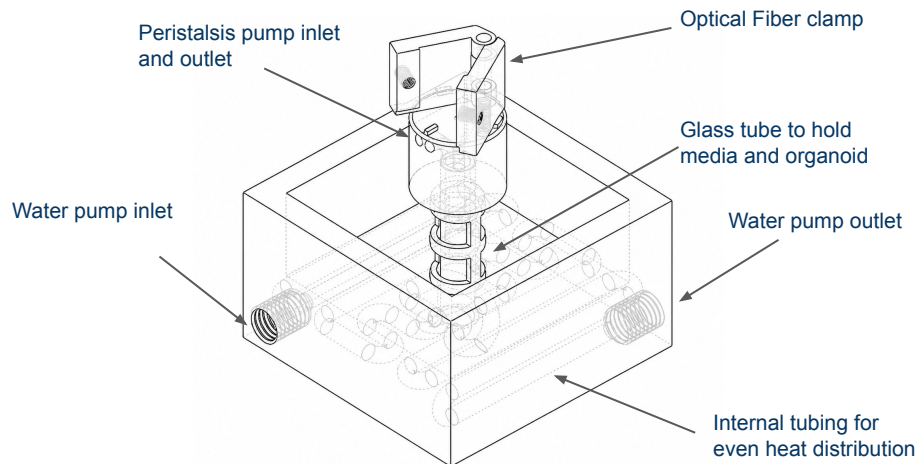
Calcium imaging confirms functional neural signaling post-incubation



Optogenetic compatibility

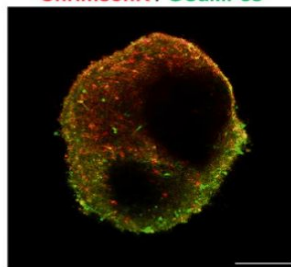
Optical fiber integrones enables precise stimulation during imaging

CAD MODEL



CALCIUM IMAGING

ChrimsonR / GCaMP8s



- CrimsonR ⇒ Red shifted opsin
- GCaMP8s ⇒ Calcium Indicator

Brain organoid expressing ChrimsonR and GCaMP8s. Optogenetic stimulation elicited calcium responses, indicating maintained neuronal function following incubation within the MRI-compatible chamber.

Design, Build, Analyze, and Learn

1. HYGIENE PRODUCT DISPENSER

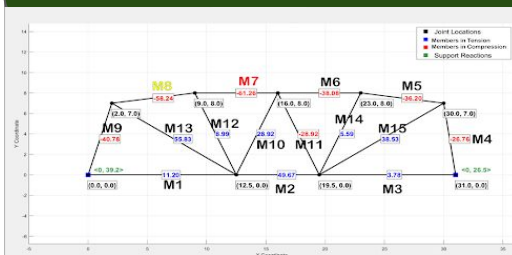
Gesture-Controlled Dispenser



- ✓ Three gesture controls and proximity control for pads, tampons, and toilet paper
- ✓ Response time: < 3 seconds
Low power
- ✓ Three stepper motors, LEDs, Arduino, and APDS 9960 gesture sensor

2. TRUSS DESIGN PROJECT

Maximizing Strength and Efficiency



- ✓ Max theoretical load: 65.698
- ✓ Critical member: #7 (7in) | Buckling strength: 61.26 oz
- ✓ Design created, tested, and validated
- ✓ Optimized geometry, even distribution of forces, and rounded top for improved performances

3. DIGITAL THERMOMETER

Real-Time Temperature Monitoring



- ✓ Temperature sensor with live readings
- ✓ LCD display for real-time monitoring
- ✓ Arduino-controlled embedded system
- ✓ Audible alarm when temperature falls outside the 68-75 degrees fahrenheit range